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| Mathematics for Machine Learning | Prof. Matthias Hein |
| Exercise Sheet 12 - 23.01.2026 | due: 27./29.01.2026 |

## Exercise 1 Coin Flip

Suppose you flip a biased coin which shows head with probability $0 < p < 1$. You predict head with probability $0 < q < 1$. you can guess correct in two different ways, sum both of the cases

a) **(2 Points)** What is the probability that your prediction is correct under the assumption that the coin flip and your prediction are independent?

b) **(2 Points)** How has $q$ to be chosen in order to maximize this probability (for fixed $p$) ?

## Exercise 2 Variance Correlation

a) **(6 Points)** Let $(X_i)_{i=1,\dots,n}$ be a set of $n$ random variables where each $X_i$ has values in $\mathbb{R}$ and has bounded variance. Show that

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \,\mathrm{Var}(X_i) + \sum_{i \neq j} a_i\, a_j \,\mathrm{Cov}(X_i, X_j).$$

$\mathrm{Var}\left(\sum a_i X_i\right) = \sum a_i^2 \mathrm{Var}(X_i)$

when X i.i.d, not this time

b) **(4 Points)** Correlation is a measure of **linear** dependence. Let $b \neq 0$ and $a \in \mathbb{R}$ and let $X$ be a random variable with values in $\mathbb{R}$ which has bounded variance. Show that $X$ and $Y = bX + a$ with $X$ taking values in $\mathbb{R}$ have correlation,

$$\mathrm{Corr}(X, Y) = \frac{b}{|b|},$$

As the absolute value of the correlation is upper bounded by one, this implies that linearly dependent random variables achieve maximal absolute correlation.

c) **(4 Points)** Show the converse, if $|\mathrm{Corr}(X, Y)| = 1$ and $X, Y$ have bounded variance and are non-constant, then there exist $a, b \in \mathbb{R}$ such that $Y = aX + b$.

## Exercise 3 $\ell_{\mathrm{inf}}$-ball equator

We consider the cube $[0,1]^d$ (this is equal to $B_\infty(\frac{1}{2}\mathbf{1}, \frac{1}{2})$) and define the "equator" to be

$$E = \left\{ x \in [0,1]^d \,\middle|\, \sum_{i=1}^{d} x_i = \frac{d}{2} \right\}.$$

We want to show that the volume of $[0,1]^d$ is concentrated around the equator.

a) **(8 Points)** Suppose that $X$ has a uniform distribution on $[0,1]^d$. Use Hoeffding inequality to show that with probability at least $1 - \frac{2}{d^2}$ it holds

$$\left| \sum_{i=1}^{d} X_i - \frac{d}{2} \right| \leq \sqrt{d \log(d)}.$$

and thus

$$\mathrm{vol}\left( \left\{ x \in [0,1]^d \,\middle|\, \left| \sum_{i=1}^{d} x_i - \frac{d}{2} \right| > \sqrt{d \log(d)} \right\} \right) \leq \frac{2}{d^2}.$$

## Exercise 4 Mixed Questions

a) **(3 Points)** Let $X \sim \mathcal{N}(0, \mathbb{1}_d)$, that is $X$ has a zero-mean Gaussian distribution in $\mathbb{R}^d$ with $\mathbb{1}_d$ being the covariance matrix (identity matrix in $\mathbb{R}^d$). Let further $A \in \mathbb{R}^{d\times d}$ be a symmetric matrix. Show that

$$\mathbb{E}[\langle X, AX\rangle] = \operatorname{tr}(A).$$

Conclude from this result further that for any matrix $B \in \mathbb{R}^{m\times d}$ it holds

$$\mathbb{E}[\|BX\|_2^2] = \mathbb{E}[\langle X, B^T BX\rangle] = \|B\|_F^2\,,$$

which yields a nice characterization of the Frobenius norm.
**Hint:** Note that $X$ is a (random) vector and not a matrix! It might be easier to write everything in terms of sums.

b) **(3 Points)** Let $X$ be an non-negative integer-valued random variable. Show that

$$\mathbb{E}[X] = \sum_{s=1}^{\infty} \mathrm{P}(X \geq s).$$

In an analogous fashion one shows for a non-negative real-valued random variable $Z$.

$$\mathbb{E}[Z] = \int_0^{\infty} \mathrm{P}(X > t)\, dt$$

**Hint:** $\mathrm{P}(X \geq s) = \sum_{k=s}^{\infty} \mathrm{P}(X = k)$ and exchange the inner and outer series.

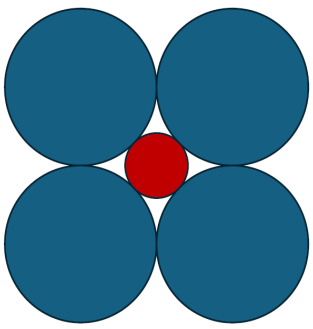
c) **(3 Points)** Let $X$ be a real-valued random variable with finite variance. Show that the difference of the median $m(X)$ of $X$ and the mean $\mathbb{E}[X]$ of $X$ can be upper-bounded as:

$$|m(X) - \mathbb{E}[X]| \;\leq\; \sqrt{\mathrm{Var}(X)}.$$

**Hint:** You may use that $m(X) = \arg\min_{c\in\mathbb{R}} \mathbb{E}[|X - c|]$ and you may use that for any convex function $f : \mathbb{R} \to \mathbb{R}$ by Jensen's inequality and any real-valued random variable $Z$:

$$f(\mathbb{E}[Z]) \leq \mathbb{E}[f(Z)].$$

d) **(3 Fun Points)** Suppose you have a box $[2, -2]^d$ and you divide each dimension into two equal halves. You get $2^d$ boxes in which you put a sphere of radius 1 which touches the box. Now you inscribe a sphere $S$ in the middle (the center is the origin) so that it just touches the other spheres (an illustration for the case $d = 2$ is given below, it shows the four blue spheres of radius one and the inner red sphere which fits inbetween them).



- What is the radius of this sphere as $d \to \infty$? (Before you start calculating: Think about the six possible solutions $\{-1, 0, \frac{1}{2}, 1, 10, \infty\}$. Suppose you have to bet on one of them. Which one would you select?)